Intro to
STAC and Cloud-Native Geo

Standards and architecture for reproducible and scalable geoscience

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Pecora 2022, Denver

Element 84

Technical Support Services Contract (TSSC)
Contractor to the USGS Earth Resources, Observation, and Science Center (EROS)

A Queryable Earth (h/t Planet)

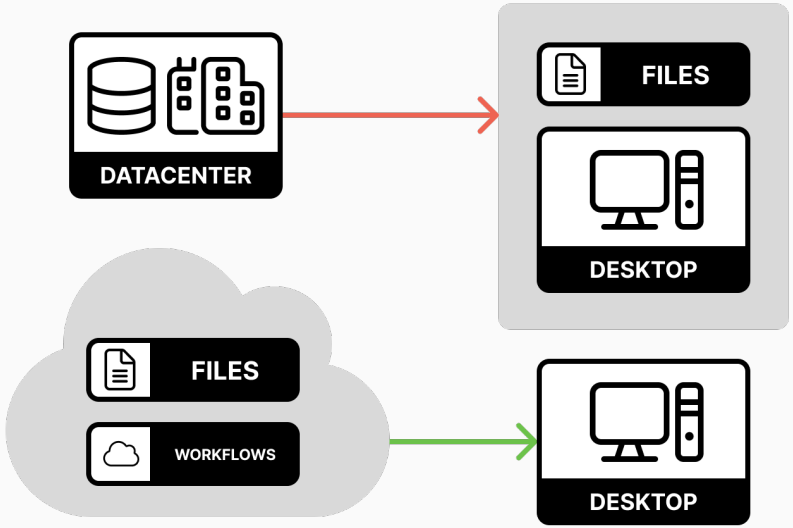
- Number of ships are in all ports in the the Mediterranean?
- How many swimming pools built in 2019 in Texas?
- Acres of forest burned in California over the last year?
- What is the average size home in Alberta?
- Total km of roads in India?
- Electrification rate of homes in Malawi?

What is Cloud-Native Geospatial?

It’s Google Earth Engine

Cloud-Native Geospatial

- Complete workflows performed in the cloud
  - Data Discovery
  - Processing
  - Analytics
  - Visualization
- via **direct data access**
  - to cloud-optimized formats
  - By users OR services
- which promotes
  - Scalability
  - Reproducibility (provenance)
  - Online Publishing (executable papers)

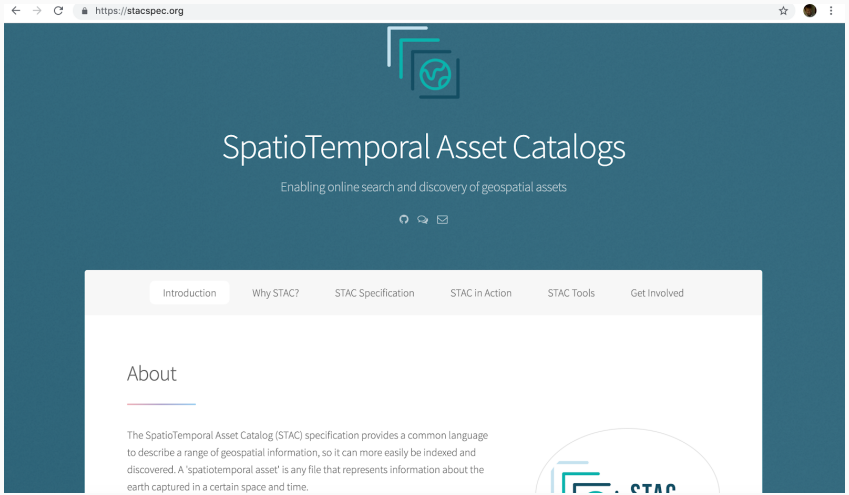


Cloud-Native Geospatial is a collection of standards and technologies for *programmatic* discovery, access, and exploitation of geospatial data
(in the cloud)

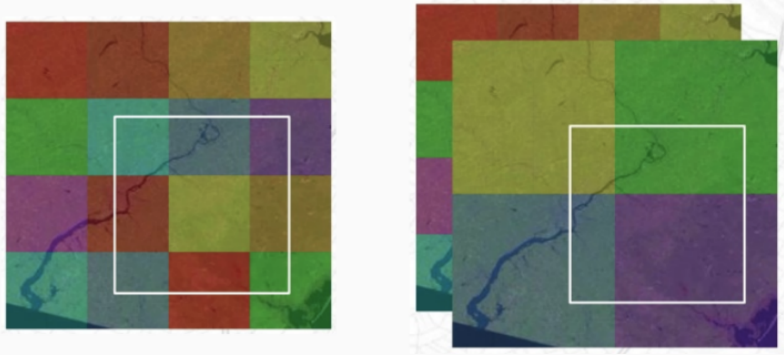
Except GEE is not..

- Open-Source
- based on standards
- deployable
- Easy to integrate with commercial providers
- Easily extensible via services

Standardized, searchable metadata



Cloud-optimized data formats



Well-characterized data



Element 84

SpatioTemporal Asset Catalogs (STAC)

AWS Open Data

About

This registry exists to help people discover and share datasets that are available via AWS resources. Learn more about sharing data on AWS.

See all usage examples for datasets listed in this registry.

See datasets from Allen Institute for Artificial Intelligence (AI2), Digital Earth Africa, Facebook Data for Good, NASA Space Act Agreement, NIH STRIDES, NOAA Big Data Program, Space Telescope Science Institute, and Amazon Sustainability Data Initiative.

Search datasets (currently 58 matching datasets)

geospatial

Add to this registry

If you want to add a dataset or example of how to use a dataset to this registry, please follow the instructions on the Registry of Open Data on AWS GitHub repository.

Unless specifically stated in the applicable dataset documentation, datasets available through the Registry of Open Data on AWS are not provided and maintained by AWS. Datasets are provided and maintained by a variety of third parties under a variety of licenses. Please check dataset licenses and related documentation to determine if a dataset may be used for your application.

Sentinel-2

agriculture | disaster response | earth observation | geospatial | natural resource | satellite imagery | sustainability

The Sentinel-2 mission is a land monitoring constellation of two satellites that provide high resolution optical imagery and provide continuity for the current SPOT and Landsat missions. The mission provides a global coverage of the Earth's land surface every 5 days, making the data of great use in on-going studies. L1C data are available from June 2015 globally. L2A data are available from September 2016 over wider Europe region and globally since January 2017.

Details →

Usage examples

- Sentinel Playground by Sinergise
- Coral-spawn slicks: Reflectance spectra and detection using optical satellite data by Hiroya Yamano, Asahi Sakuma, Saki Harii
- Spectator - tracking Sentinel 2, accessing the data and quick preview by Spectator
- Tutorial for using Sentinel-2 data by Antti Lipponen
- FME Landsat-8/Sentinel-2 File Selector by Safe Software

See 20 usage examples →

USGS Landsat

agriculture | disaster response | earth observation | geospatial | natural resource | satellite imagery | sustainability

This joint NASA/USGS program provides the longest continuous space-based record of Earth's land in existence. Every day, Landsat satellites provide essential information to help land managers and policy makers make wise decisions about our resources and our environment. Data is provided for Landsats 1, 2, 3, 4, 5, 7, and 8.

Details →

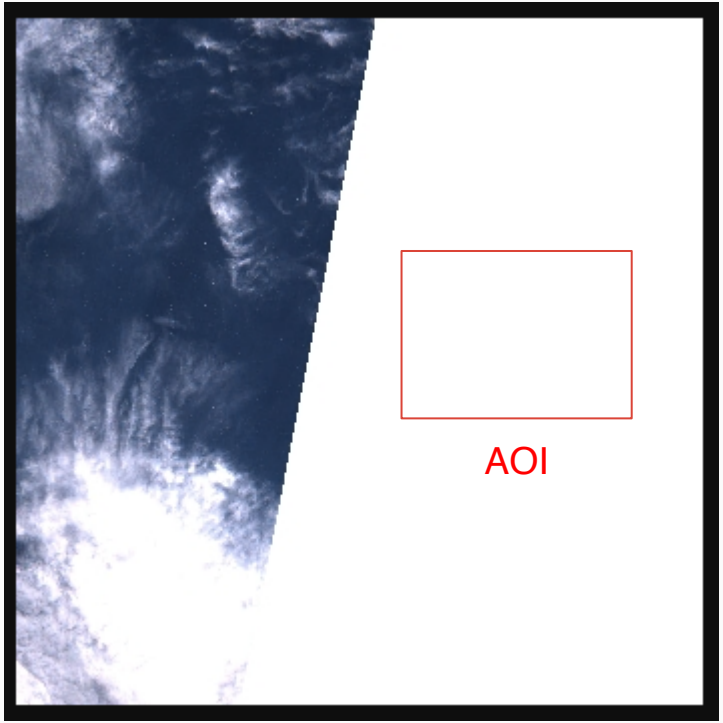
Usage examples

- Apps for exploring and analyzing Landsat imagery on the fly by Esri
- FME Landsat-8/Sentinel-2 File Selector by Safe Software
- Cloud Segmentation in Landsat-8 Images by Ingrid Grenet
- Landsat in the Cloud – Supporting Current Workflows by USGS
- Development Seed Geolambda by Matthew Hanson

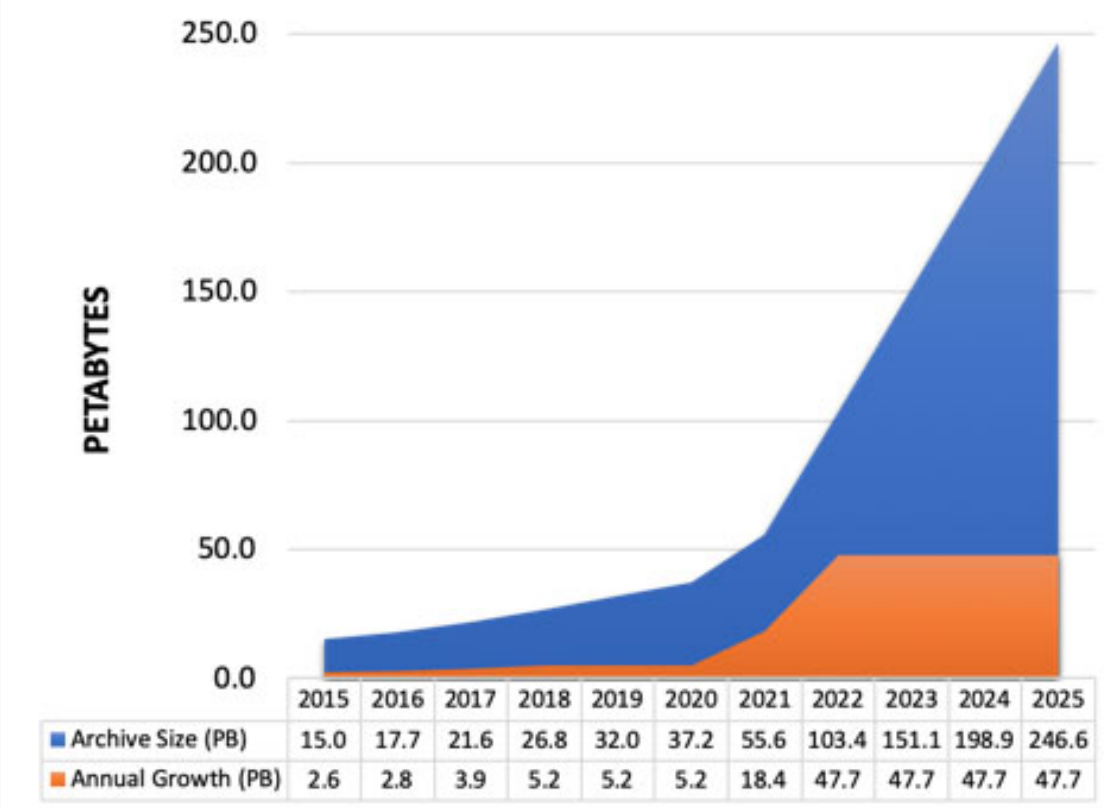
Discovery via filename

```
matthew@aeon:~$ aws s3 ls s3://sentinel-s2-l2a/tiles/59/M/RT/2020/6/11 --request-payer
 --recursive | more
```

Tile Geometry vs Data Geometry



NASA Archive

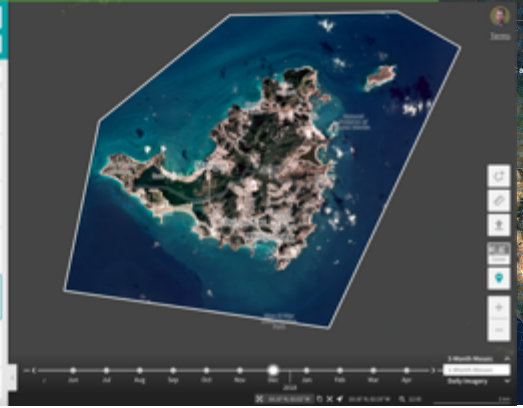


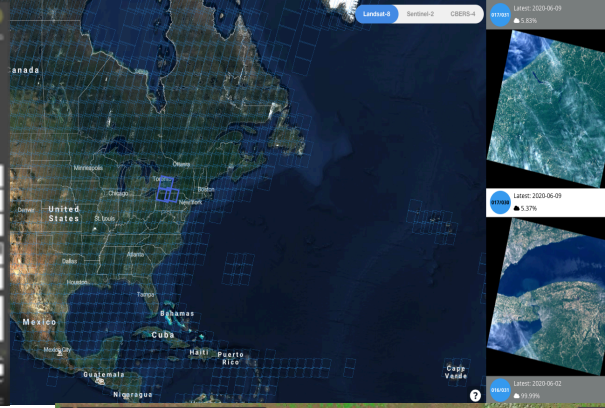
	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Archive Size (PB)	15.0	17.7	21.6	26.8	32.0	37.2	55.6	103.4	151.1	198.9	246.6
Annual Growth (PB)	2.6	2.8	3.9	5.2	5.2	5.2	18.4	47.7	47.7	47.7	47.7

If your metadata is not indexed your data does not exist

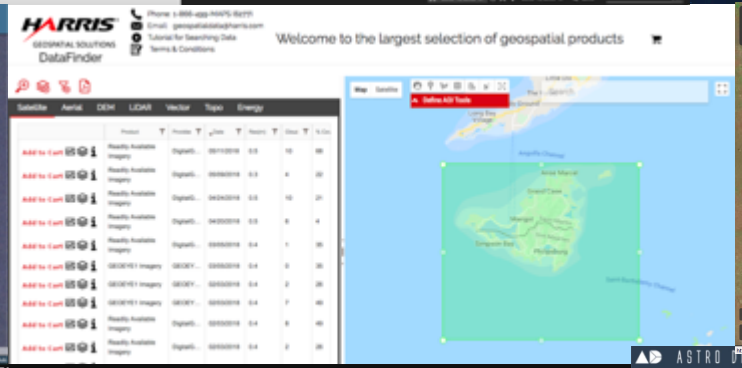

Saint Martin
Area of Interest
Filters
Source
Collected
Area Clouds
Area Nadir
Actions

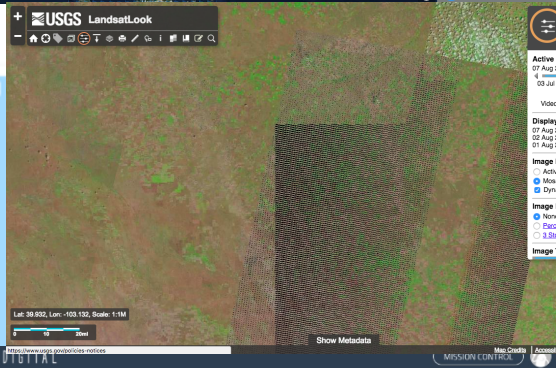

Sint Maarten, Netherlands
Browse
Compare
Timelapse
1-Month Mosaic
April 2018
March 2018
February 2018
January 2018
December 2017
November 2017
October 2017

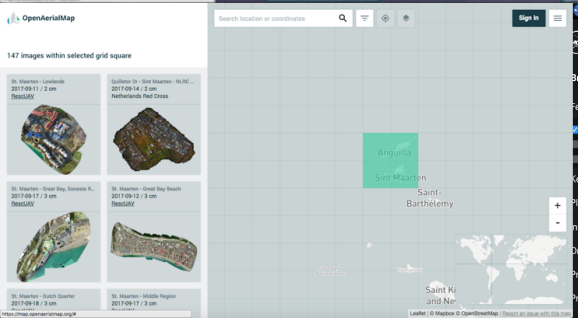


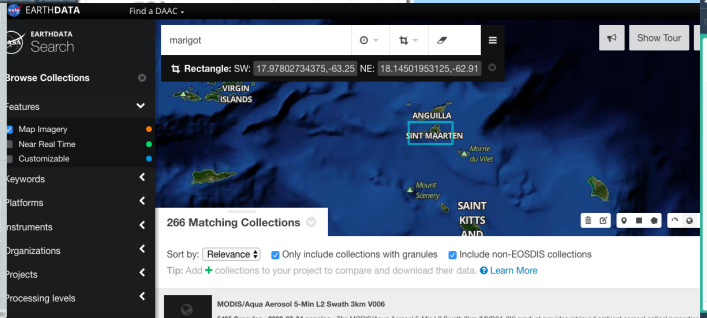

Landsat-8
Sentinel-2
CBERS-4
United States
México
Cuba
Bahamas
Haiti
Puerto Rico
Guatemala
Nicaragua
Cape Verde

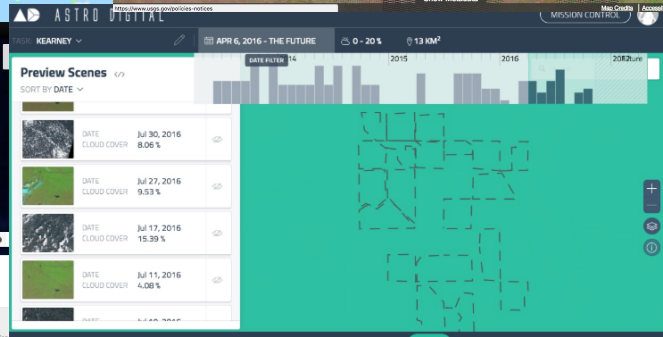

SENTINEL Hub Playground
Rendering
Effects
Natural color
Color Infrared (vegetation)
False color (urban)
Agriculture
Vegetation Index
Moisture Index
Geology
Bathymetric
Atmospheric penetration
GENERATE


HARRIS GEOSPATIAL SOLUTIONS
DataFinder
Welcome to the largest selection of geospatial products
Satellite
Aerial
DEM
LiDAR
Vector
Topo
Energy


USGS LandsatLook
Lat: 39.932, Lon: -103.132, Scale: 1:1M
Show Metadata

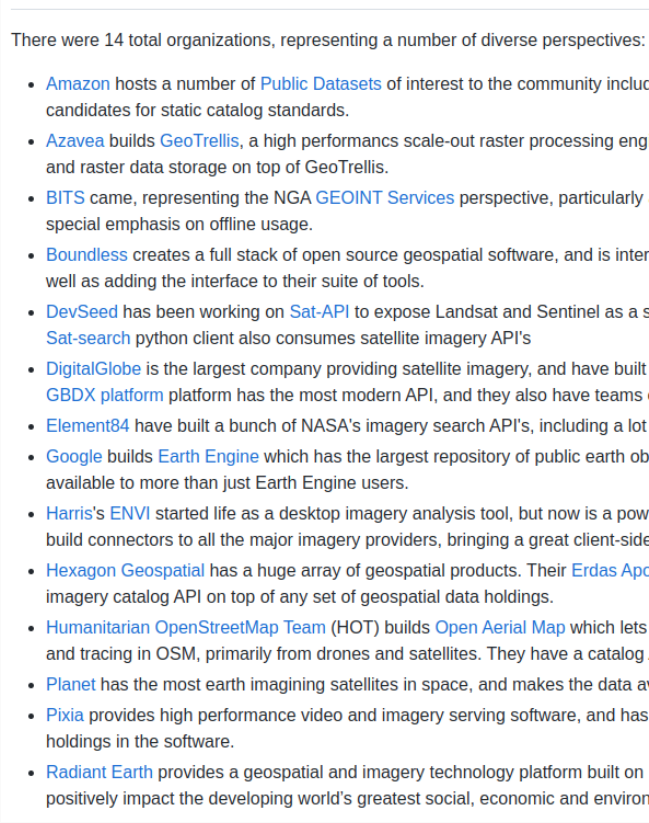

OpenAerialMap
147 images within selected grid square


EARTHDATA
Search
Find a DAAC
Browse Collections
Features
Map Imagery
Near Real Time
Customizable
Keywords
Platforms
Instruments
Organizations
Projects
Processing levels
marigot
Show Tour
Rectangle: SW: 17.97802734375,-63.25 NE: 18.14501953125,-62.91
VIRGIN ISLANDS
ANGUILLA
SINT MAARTEN
SAINT KITTS AND
266 Matching Collections
Sort by: Relevance
Only include collections with granules
Include non-EOSDIS collections
Tip: Add collections to your project to compare and download their data.
Learn More
MODIS/Aqua Aerosol 5-Min L2 Swath 3km V006


ASTRO DIGITAL
MISSION CONTROL
KEARNEY
APR 6, 2016 - THE FUTURE
0 - 20 %
13 KM²
Preview Scenes
SORT BY DATE
DATE FILTER
DATE Jul 30, 2016
CLOUD COVER 8.06 %
DATE Jul 27, 2016
CLOUD COVER 9.53 %
DATE Jul 17, 2016
CLOUD COVER 15.39 %
DATE Jul 11, 2016
CLOUD COVER 4.08 %

“Imagery Metadata Spec”

- Boulder, October 2017
- Requirements
  - User-centric search and discovery
  - Leverage other standards (GeoJSON)
  - Not a replacement for existing metadata
  - Crawlable (therefore indexable)
  - Simple extensible core
- Development practices
  - Release early, release often
  - Implement data and software
  - Iterate

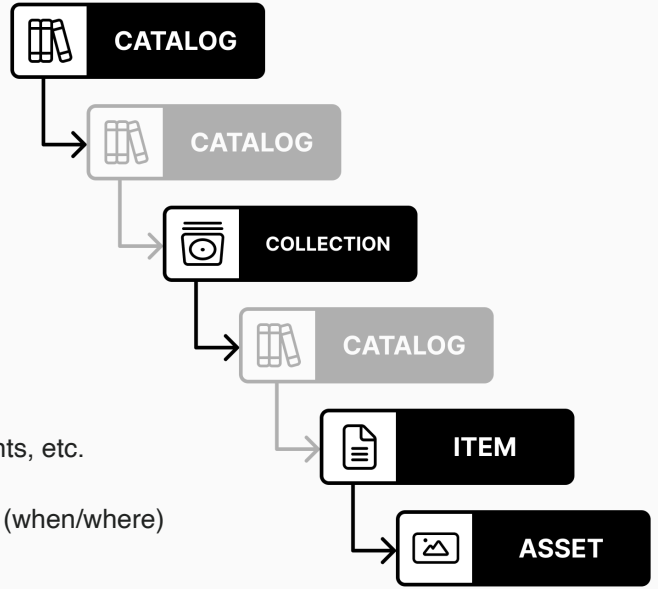


There were 14 total organizations, representing a number of diverse perspectives:

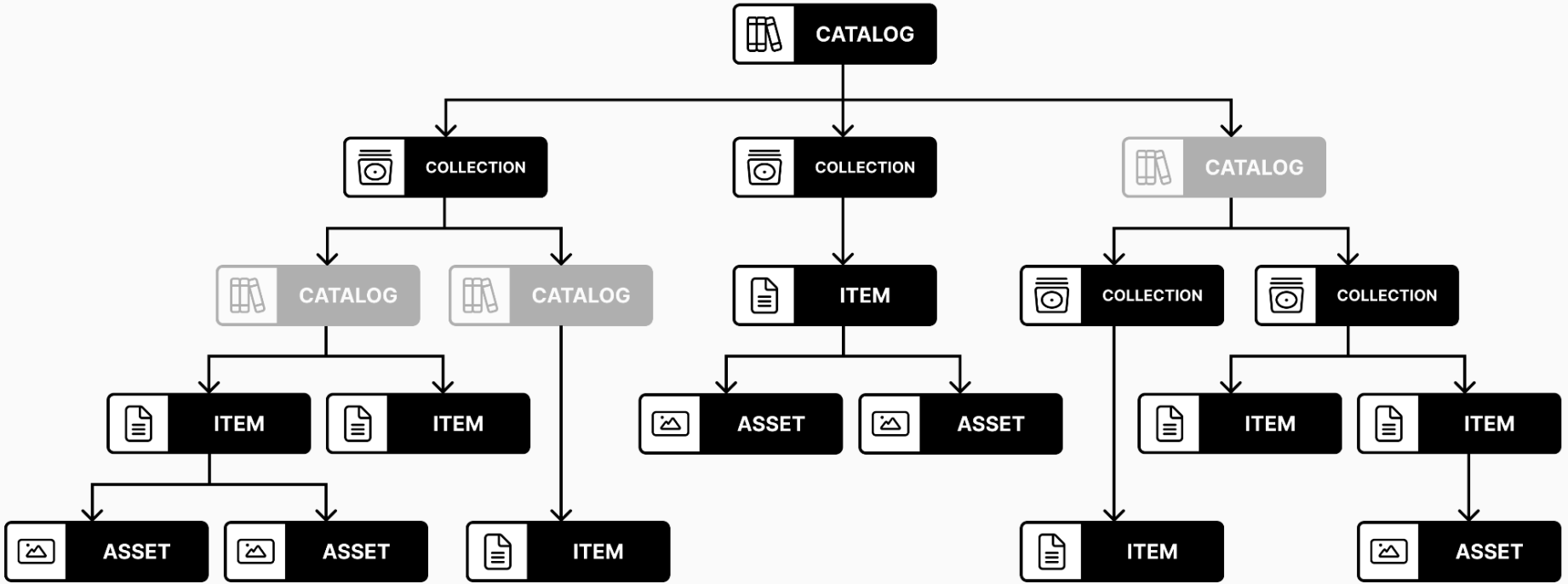
- Amazon hosts a number of Public Datasets of interest to the community includ candidates for static catalog standards.
- Azavea builds GeoTrellis, a high performancs scale-out raster processing eng and raster data storage on top of GeoTrellis.
- BITS came, representing the NGA GEOINT Services perspective, particularly special emphasis on offline usage.
- Boundless creates a full stack of open source geospatial software, and is inter well as adding the interface to their suite of tools.
- DevSeed has been working on Sat-API to expose Landsat and Sentinel as a s Sat-search python client also consumes satellite imagery API's
- DigitalGlobe is the largest company providing satellite imagery, and have built GBDX platform platform has the most modern API, and they also have teams
- Element84 have built a bunch of NASA's imagery search API's, including a lot
- Google builds Earth Engine which has the largest repository of public earth ob available to more than just Earth Engine users.
- Harris's ENVI started life as a desktop imagery analysis tool, but now is a pow build connectors to all the major imagery providers, bringing a great client-side
- Hexagon Geospatial has a huge array of geospatial products. Their Erdas Apo imagery catalog API on top of any set of geospatial data holdings.
- Humanitarian OpenStreetMap Team (HOT) builds Open Aerial Map which lets and tracing in OSM, primarily from drones and satellites. They have a catalog
- Planet has the most earth imagining satellites in space, and makes the data av
- Pixia provides high performance video and imagery serving software, and has holdings in the software.
- Radiant Earth provides a geospatial and imagery technology platform built on positively impact the developing world's greatest social, economic and environ

SpatioTemporal Asset Catalogs (STAC)

- Specifications to describe geospatial data for search, discovery, and use in the cloud
  - JSON
  - Crawlable (indexable)
  - Standardize metadata fields
  - Standard API for querying (OGC API Features extension)



- Catalogs
  - Container that links to other catalogs and collections
  - Top-level catalog is the Root Catalog
- Collections
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Items
  - A single scene or set of datafiles for a specific location at a specific date and time (when/where)


CATALOG
COLLECTION
COLLECTION
CATALOG
CATALOG
CATALOG
ITEM
COLLECTION
COLLECTION
ITEM
ITEM
ASSET
ASSET
ITEM
ITEM
ASSET
ASSET
ITEM
ITEM
ASSET

The STAC specifications describe describe geospatial data (stac-spec), and how to searching that data (stac-api-spec)

STAC Status

stac-spec
1.0.0
(May 25, 2021)

OS Projects
30+

stac-api-spec
1.0.0-rc.1
(March 17, 2022)

Extensions
30+

Contributors
50+
(19 w/ 10+ commits)

Public Catalogs
32+
w/ 1+ billion Items

A simple core spec

Field Name	Type	Description
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
stac_version	string	**REQUIRED.** The STAC version the Item implements.
stac_extensions	[string]	A list of extensions the Item implements.
id	string	**REQUIRED.** Provider identifier. The ID should be unique within the Collection that contains the Item.
geometry	GeoJSON Geometry Object | null	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Coordinates are specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84.
bbox	[number]	**REQUIRED if `geometry` is not `null`.** Bounding Box of the asset represented by this Item, formatted according to RFC 7946, section 5.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the Item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is strongly recommended.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type). This field is *required* if such a relation type is present and is *not allowed* otherwise. This field provides an easy way for a user to search for any Items that belong in a specified Collection. Must be a non-empty string.

stac-extensions.github.io

STAC Extensions

The stac-extensions github organization is a home for extensions to the SpatioTemporal Asset Catalog specification.

To learn about STAC and Extensions start with the extensions section of the core specification. It explains how extensions work, lists all the known extensions, and has instructions for how to go about 'extending STAC'.

The stac-extensions github organization is a home for many of the leading 'community extensions', providing a neutral home for collaboration. Many of these used to be in the core stac-spec repository, but were removed for 1.0.0 so they could evolve at their own pace, instead of having to follow the core STAC release cycle.

List of STAC Extensions

The definitive list of STAC Extensions is in the core spec repository, in the Extensions README. This includes the official core extensions, as well as a complete list of 'community' extensions. A subset of the community extensions use this stac-extensions GitHub organization, and those are listed in the next section.

Extensions in stac-extensions organization

These extensions add new fields or semantics to STAC objects.

Extension Title	Field Name Prefix	Scope	Maturity	Published?	Description
CARD4L	card4l.	Item	*Proposal*	No	How to comply to the CEOS CARD4L product family specifications (Optical and SAR)
Data Cube	cube	Item, Collection	*Proposal*	Yes	Data Cube related metadata, especially to describe their dimensions

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Notable Extensions

projection

Field Name	Type	Description
proj:epsg	integer|null	**REQUIRED.** EPSG code of the datasource
proj:wkt2	string|null	WKT2 string representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:projjson	PROJJSON Object|null	PROJJSON object representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:geometry	GeoJSON Geometry Object	Defines the footprint of this Item.
proj:bbox	[number]	Bounding box of the Item in the asset CRS in 2 or 3 dimensions.
proj:centroid	Centroid Object	Coordinates representing the centroid of the Item (in lat/long)
proj:shape	[integer]	Number of pixels in Y and X directions for the default grid
proj:transform	[number]	The affine transformation coefficients for the default grid

Electro-Optical

Field Name	Type	Description
eo:bands	[Band Object]	An array of available bands where each object is a Band Object. If given, requires at least one band.
eo:cloud_cover	number	Estimate of cloud cover

SAR

Field Name	Type	Description
sar:instrument_mode	string	**REQUIRED.** The name of the sensor acquisition mode that is commonly used. This should be the short name, if available. For example, `WV` for "Wave mode" of Sentinel-1 and Envisat ASAR satellites.
sar:frequency_band	string	**REQUIRED.** The common name for the frequency band to make it easier to search for bands across instruments. See section "Common Frequency Band Names" for a list of accepted names.
sar:center_frequency	number	The center frequency of the instrument, in gigahertz (GHz).
sar:polarizations	[string]	**REQUIRED.** Any combination of polarizations.
sar:product_type	string	**REQUIRED.** The product type, for example `SSC`, `MGD`, or `SGC`
sar:resolution_range	number	The range resolution, which is the maximum ability to distinguish two adjacent targets perpendicular to the flight path, in meters (m).
sar:resolution_azimuth	number	The azimuth resolution, which is the maximum ability to distinguish two adjacent targets parallel to the flight path, in meters (m).
sar:pixel_spacing_range	number	The range pixel spacing, which is the distance between adjacent pixels perpendicular to the flight path, in meters (m). Strongly RECOMMENDED to be specified for products of type `GRD`.
sar:pixel_spacing_azimuth	number	The azimuth pixel spacing, which is the distance between adjacent pixels parallel to the flight path, in meters (m). Strongly

raster

Field Name	Type	Description
nodata	number|string	Pixel values used to identify pixels that are nodata in the band either by the pixel value as a number or `nan`, `inf` or `-inf` (all strings).
sampling	string	One of `area` or `point`. Indicates whether a pixel value should be assumed to represent a sampling over the region of the pixel or a point sample at the center of the pixel.
data_type	string	The data type of the pixels in the band. One of the data types as described above.
bits_per_sample	number	The actual number of bits used for this band. Normally only present when the number of bits is non-standard for the `datatype`, such as when a 1 bit TIFF is represented as byte
spatial_resolution	number	Average spatial resolution (in meters) of the pixels in the band.
statistics	Statistics Object	Statistics of all the pixels in the band

OGC API (New)	OGC Legacy (Old)	What is it for?
Common	OWS Common	Common API elements
Features	Web Feature Service (WFS)	Serving geospatial vector data
Coverages	Web Coverage Services (WCS)	Serving geospatial raster data
Maps	Web Map Service (WMS)	Serving rendered map layers
Tiles	Web Map Tile Service (WMTS)	Serving data as tiles
Processes	Web Processing Service (WPS)	Invoking geospatial processing requests
Records	Catalog Service for the Web (CSW)	Searching metadata

STAC API

STAC API spec
https://github.com/radiantearth/stac-api-spec

OGC API: Features
+
STAC Extensions

OpenAPI Doc: https://stacspec.org/STAC-ext-api.html

OGC API - Features Endpoints

The core OGC API - Features endpoints are shown below, with details provided in an OpenAPI specification document.

Endpoint	Returns	Description
`/`	JSON	Landing page, links to API capabilities
`/conformance`	JSON	Info about standards to which the API conforms
`/collections`	JSON	Object with a list of Collections contained in the catalog and links
`/collections/{collectionId}`	Collection	Returns single Collection JSON
`/collections/{collectionId}/items`	ItemCollection	GeoJSON FeatureCollection-conformant entity of Items in collection
`/collections/{collectionId}/items/{featureId}`	Item	Returns single Item (GeoJSON Feature)

STAC Endpoints

STAC provides some additional endpoints for the root Catalog itself, as well as the capability to search the Catalog. Note that a STAC API does not need to implement OAFeat, in which case it would only support the endpoints given below. See the OpenAPI specification document.

Endpoint	Returns	Description
`/`	Catalog	Extends `/` from OAFeat to return a full STAC catalog.
`/search`	ItemCollection	Retrieves a group of Items matching the provided search predicates, probably containing search metadata from the `search` extension

Search endpoints

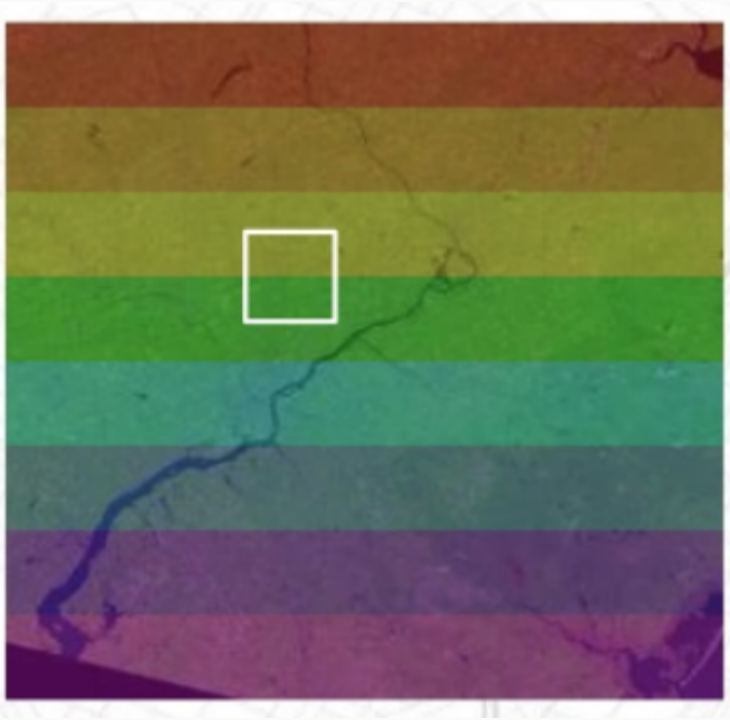
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Cloud-Optimized File Formats

Cloud Native Data Formats

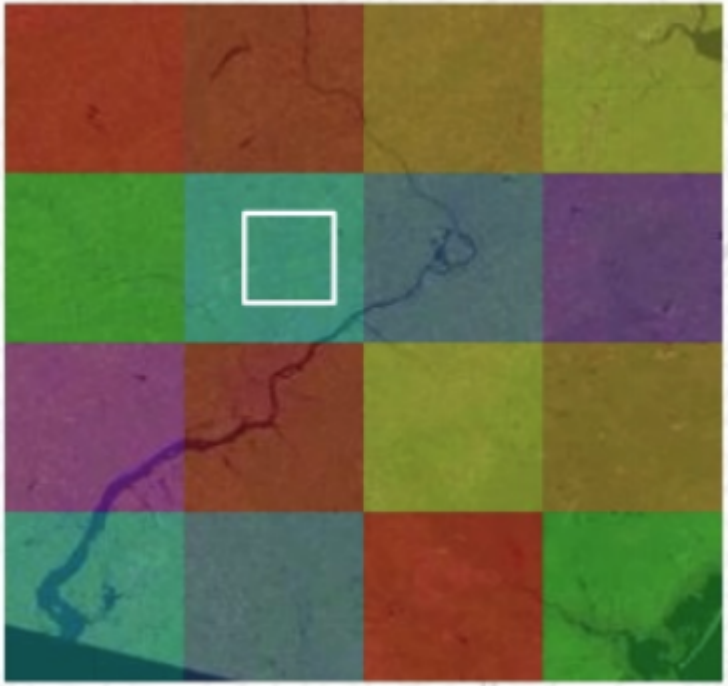
Format	Data Type
Cloud-Optimized GeoTIFF (COG)	Raster (good for EO data)
Zarr	Raster (good for modeled data)
Cloud-Optimized Point Cloud (COPC)	Point Clouds
GeoParquet	Vector

Striped Chunks





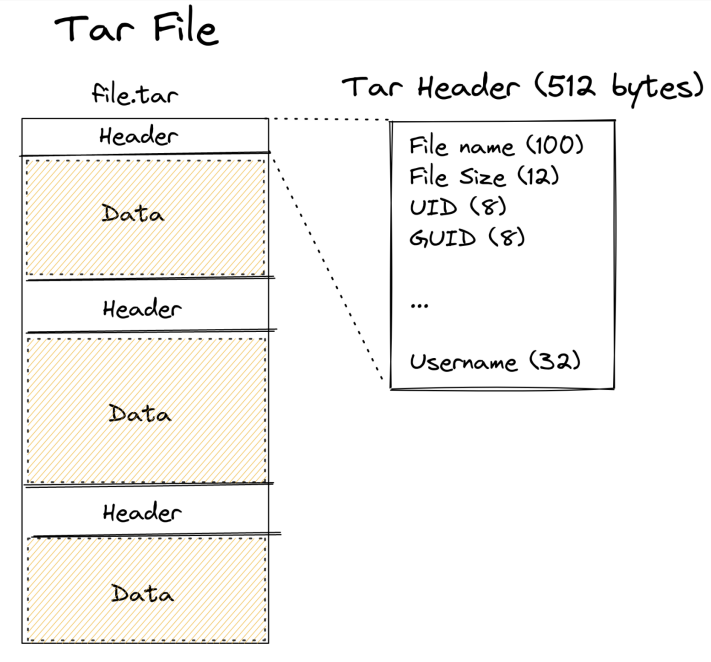
Tiled Chunks

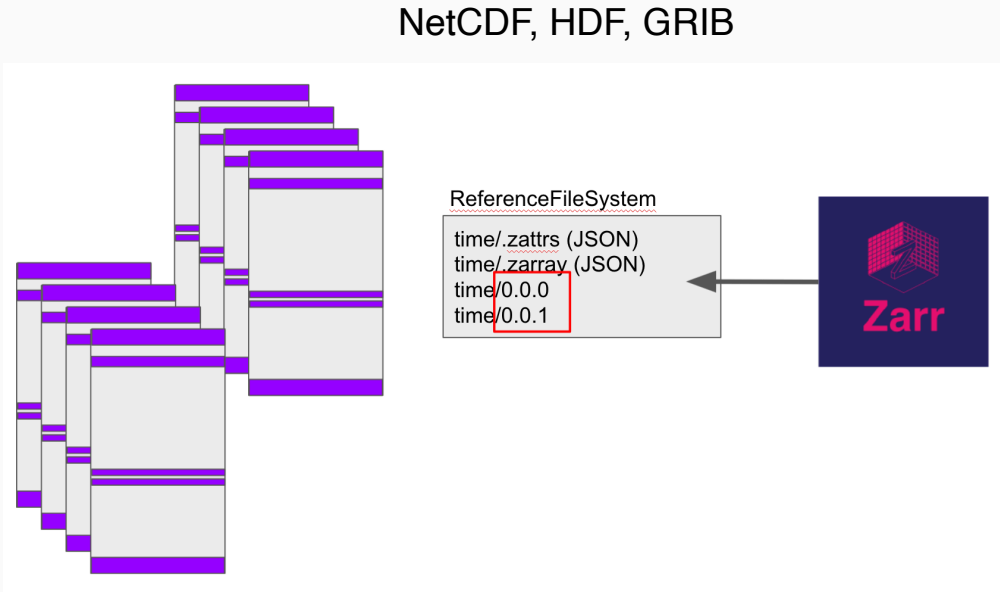




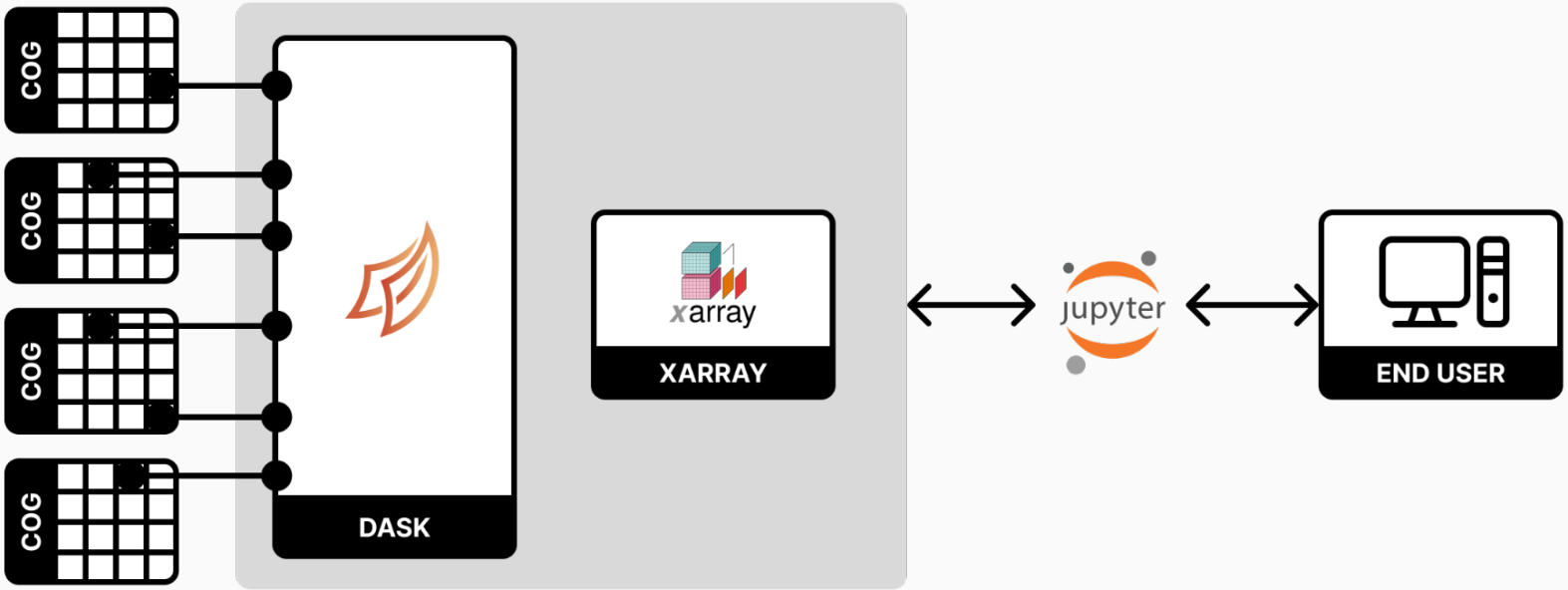
A Cloud Native Dataset is one with small addressable chunks (via files, internal tiles, or both)

Cloud-Optimized <insert-legacy-format>?

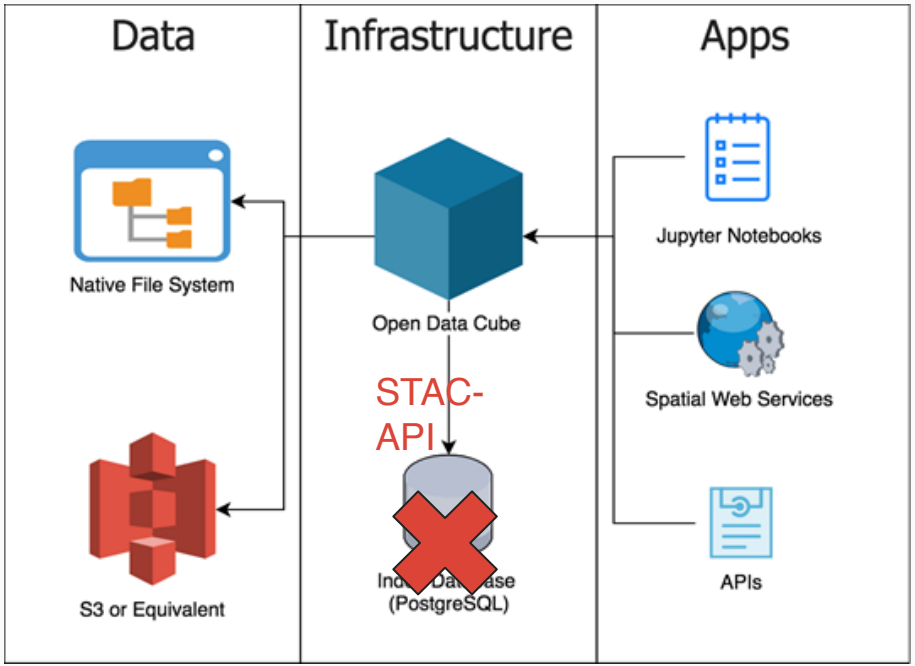




https://github.com/linz/cotar


COG
COG
COG
COG
DASK
xarray
XARRAY
jupyter
END USER

ODC-STAC (OpenDataCube)



Element 84

Australian Government
Geoscience Australia

Digital Earth
AFRICA

Find STAC Items to Load

[3]:
```
km2deg = 1.0 / 111
x, y = (113.887, -25.843)  # Center point of a query
r = 100 * km2deg

query = catalog.search(
    collections=["sentinel-s2-l2a-cogs"],
    datetime="2021-09-16",
    limit=10,
    bbox=(x - r, y - r, x + r, y + r),
)

items = list(query.get_items())
print(f"Found: {len(items):d} datasets")
```

```
Found: 9 datasets
```

Construct Dask Dataset

Note that even though there are 9 STAC Items on input, there is only one timeslice on output. This is because of `groupy="solar_day"`. With that setting `stac_load` will place all items that occured on the same day (as adjusted for the timezone) into one image plane.

[4]:
```
# crs = "epsg:32749"  # native UTM projection in the query region
crs = "epsg:3857"  # Since we will plot it on a map we need to use 3857 projection
zoom = 2 ** 5  # overview level 5

xx = stac_load(
    items,
    output_crs=crs,
    resolution=(-10 * zoom, 10 * zoom),
    chunks={},
    groupby="solar_day",
    measurements=["red", "green", "blue"],
    stac_cfg=cfg,
)
xx
```

[4]:

xarray.Dataset

▸ Dimensions: (**time**: 1, **y**: 1098, **x**: 833)

Dynamic analytics should start with a query

Find STAC Items to Load

[3]:
```
km2deg = 1.0 / 111
x, y = (113.887, -25.843)  # Center point of a query
r = 100 * km2deg

query = catalog.search(
    collections=["sentinel-s2-l2a-cogs"],
    datetime="2021-09-16",
    limit=10,
    bbox=(x - r, y - r, x + r, y + r),
)

items = list(query.get_items())
print(f"Found: {len(items):d} datasets")
```

```
Found: 9 datasets
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Construct Dask Dataset

Note that even though there are 9 STAC Items on input, there is only one timeslice on output. This is because of `groupy="solar_day"`. With that setting `stac_load` will place all items that occured on the same day (as adjusted for the timezone) into one image plane.

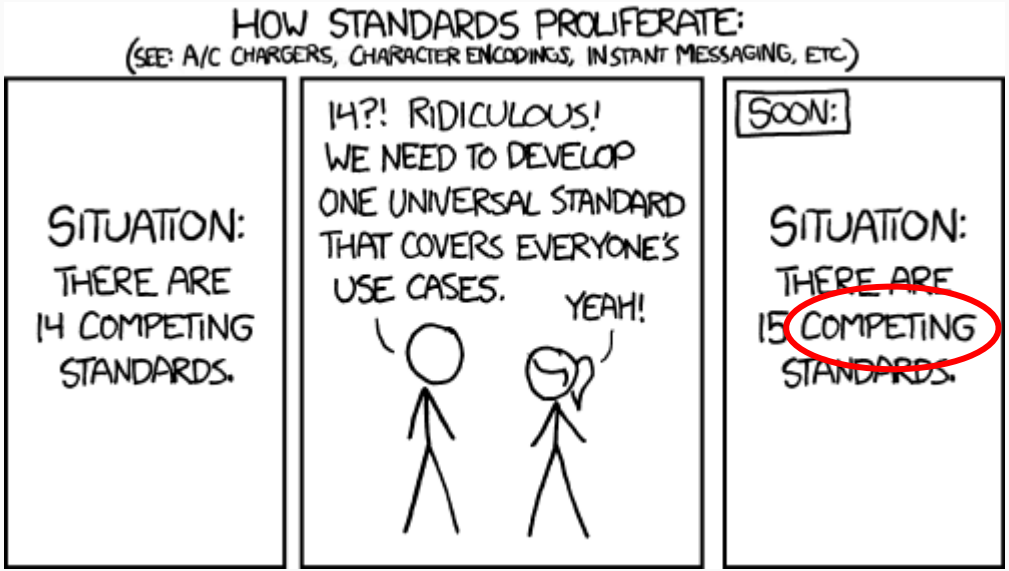
[4]:
```
# crs = "epsg:32749"  # native UTM projection in the query region
crs = "epsg:3857"  # Since we will plot it on a map we need to use 3857 projection
zoom = 2 ** 5  # overview level 5

xx = stac_load(
    items,
    output_crs=crs,
    resolution=(-10 * zoom, 10 * zoom),
    chunks={},
    groupby="solar_day",
    measurements=["red", "green", "blue"],
    stac_cfg=cfg,
)
xx
```

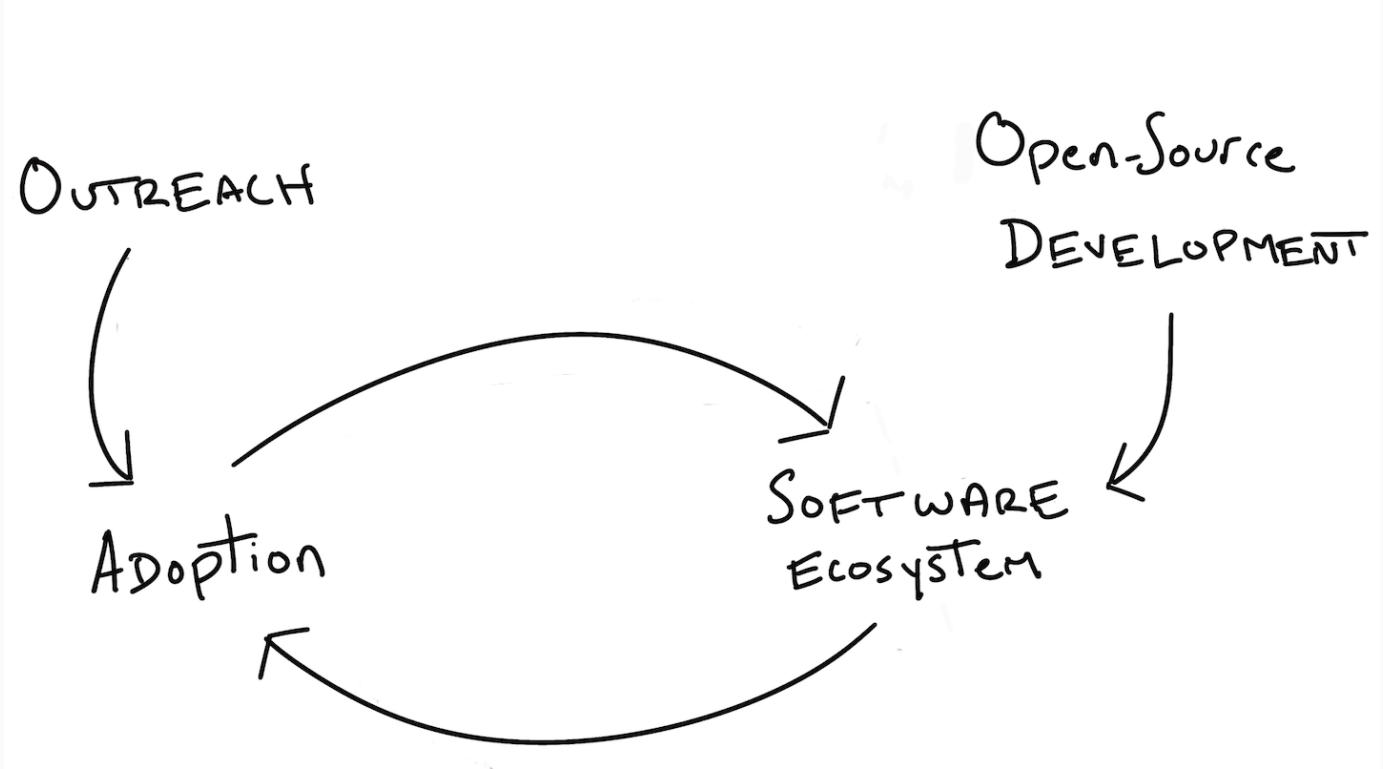
[4]:
xarray.Dataset

▸ Dimensions: (**time**: 1, **y**: 1098, **x**: 833)

Adoption


HOW STANDARDS PROLIFERATE:
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC.)
SITUATION: THERE ARE 14 COMPETING STANDARDS.
14?! RIDICULOUS! WE NEED TO DEVELOP ONE UNIVERSAL STANDARD THAT COVERS EVERYONE'S USE CASES.
YEAH!
SOON:
SITUATION: THERE ARE 15 COMPETING STANDARDS.

xkcd


OUTREACH
Open-Source
DEVELOPMENT
ADOPTION
SOFTWARE
ECOSYSTEM

Sponsors


planet.
Radiant Earth Foundation
EARTH IMAGERY FOR IMPACT
Microsoft
geo
Element 84
MAXAR TECHNOLOGIES
ARTURO
USGS
science for a changing world
cosmiQ works
aws
Digital Earth AFRICA
WORLD BANK GROUP
azavea
NASA
UMBRA
open EO
PIXEL8
development SEED
astræa
NEAR SPACE LABS
THE CLIMATE CORPORATION
VULCAN Technology
L3HARRIS
Google Earth Engine

Chris Holmes @opencholmes · 14m
Replying to @GeoSkeptic
@planet uses @STACspec to distribute some bits of open data, we provide a stactools plugin for python and CLI to convert Planet Orders to STAC, and STAC will likely play a very prominent role in our next generation of API's.

Subit Chakrabarti @abitmore · 4h
Replying to @GeoSkeptic and @opencholmes
At @Cloud2Street we heavily really on STAC for satellite data ingestion and index all the flood maps we produce on a STAC server !
1 4

Scott Soenen @ScottSoenen · 19h
Replying to @GeoSkeptic and @opencholmes
Hydrosat is building on #STAC and plans to make data products available through a STAC API. We're grateful to have shoulders of giants on which to stand.
1 9

Capella Space @capellaspace · 18h
Replying to @GeoSkeptic and @opencholmes
At Capella we use #STAC as a fundamental component of our data access and delivery systems
1 5

Burke Greer @bgr33r · Sep 28
Replying to @GeoSkeptic and @opencholmes
We've got it running in our backend with marblecutter ready for ingest/display at @GeositeInc. This means we're interoperable against any STAC-friendly sources.
2 9

Ben Tuttle
@bentuttle1
Replying to @GeoSkeptic and @opencholmes
STAC is at the core of our geospatial services at @arturo_ai and we will continue to leverage and contribute to it as we develop new capabilities (both internal and customer facing).

Joe Morrison @mouthofmorrison · 21h
Replying to @GeoSkeptic and @opencholmes
We're building our archive on top of STAC and even exploring making our tasking interface as STAC-like as possible. @npcasler and @notthatbreezy would know better than me.
1 5

Rema @nearspacerema · Sep 28
Replying to @GeoSkeptic @HamedAlemo and @opencholmes
Yep, we do @NearSpaceLabs
1 6

James O'Connor @James_O_Connor1 · Sep 28
Replying to @GeoSkeptic and @opencholmes
At @satellitevu we have a stac-fastapi deployment running internally for our flight trial data and are planning a release of a publically accessible static catalog of sample imagery. We're prelaunch still, but absolutely going to use it in the future 😎
1 2 10

Joe Morrison @mouthofmorrison · 21h
nice!
2

Jan Žák @zakjan · 12h
Replying to @GeoSkeptic and @opencholmes
@WeatherLayers client API is based on STAC with additional fields to describe default visualization config (raster, contours, highs/lows, particle animation). Raster part is similar to 'color palette' from the closed 'rendering hints' PR, but there is more

> **Chris Holmes** @opencholmes · Aug 30
> Replying to @zakjan @DanielJDufour and 3 others
> Yes, that's a good place to start. It may end making sense to be its own related extension. I also had a never finished 'rendering hints' extension that could fit this. Most of the fields there ended up making more sense in raster, so not much was left to push forward.

2

Marc Pfister @marcpfister · 20h
Replying to @GeoSkeptic and @opencholmes
At Maxar, STAC files are a core part of our ARD product: ard.maxar.com/docs/about/#im.... They're the peanut butter to the COG jelly.
1 1 9

Alex Leith @alexgleith · 15h
haha, love it!

Although I think I'd prefer the STAC sauce to the COG snag!
1 1

Show replies

Zac Deziel @zacdezgeo · 14h
Replying to @GeoSkeptic and @opencholmes
Fastmap, a Calgary based startup, is building their customer facing API based on STAC
1 4

Jason Suwala @jsuwala · Sep 28
Replying to @GeoSkeptic @Element84 and @opencholmes
Yup - at @HatfieldConsult we're using STAC for what we're doing at geoanalytics.ca and

Element 84

Public STAC APIs

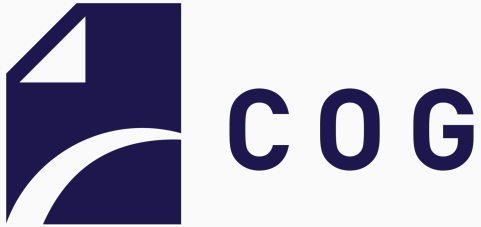
- USGS Landsat (official)
  - https://landsatlook.usgs.gov/stac-server
- Earth-Search (Sentinel-2 on AWS)
  - https://earth-search.aws.element84.com/v0
- NASA Common Metadata Repository (CMR) STAC proxy
  - https://cmr.earthdata.nasa.gov/stac
- Microsoft Planetary Computer
  - https://planetarycomputer.microsoft.com/api/stac/v1
- Google Earth Engine
  - https://earthengine-stac.storage.googleapis.com/catalog/catalog.json
- See STAC Index for more
  - https://stacindex.org/


Landsat Program



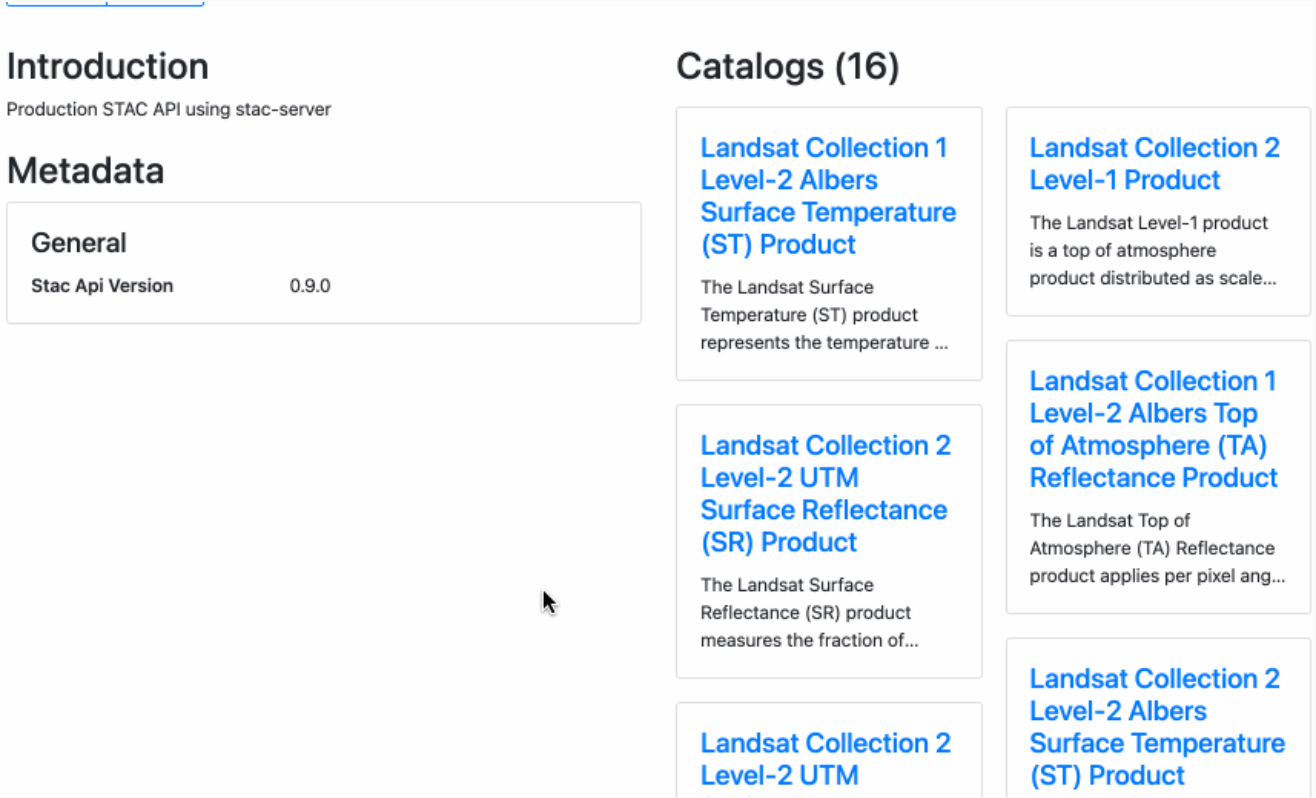

STAC

+


COG

Landsat Collection 2

https://landsatlook.usgs.gov/stac-server



Landsat Collection2 Collections



ID	Level	CRS	Description
landsat-c2l1	1	UTM	TOA, Bright Temperature (optical - LWIR)
landsat-c2l2-sr	2	UTM	Surface reflectance (optical - MWIR)
landsat-c2l2-st	2	UTM	Surface temperature (LWIR)
landsat-c2l2alb-sr*	2	Albers	Surface reflectance, Landsat path/row scenes
landsat-c2l2alb-st*	2	Albers	Surface temperature, Landsat path/row scenes
landsat-c2ard-sr*	2	Albers	Surface reflectance, tiled
landsat-c2ard-st*	2	Albers	Surface temperature, tiled
landsat-c2l3-ba	3	Albers	Burned area
landsat-c2l3-dswe	3	Albers	Dynamic Surface Water Extent
landsat-c2l3-fsca	3	Albers	Fractional Snow Covered Area

* Collections for non-atmospherically corrected versions as well

Software Ecosystem

stac-utils

Overview | Repositories | Packages | People | Teams | Projects | Settings

Find a repository... | Type | Language | Sort | New repository

pystac Public
Python library for working with any SpatioTemporal Asset Catalog (STAC)
Python 58 139 85 (6 issues need help) 1 Updated 7 minutes ago

stac-layer Public
Visualize a STAC Item or Collection on a Leaflet Map
HTML CC0-1.0 0 1 4 2 Updated 22 minutes ago

stac-fastapi Public
STAC API implementation with FastAPI.
Python MIT 31 71 59 5 Updated 6 hours ago

stac-vrt Public
Python MIT 2 18 5 1 Updated yesterday

pystac-client Public
Python client for STAC Catalogs and APIs
Python 9 40 8 0 Updated 2 days ago

pgstac Public
Schema, functions and a python library for storing and accessing STAC collections and items in PostgreSQL
sql postgis stac
PLpgSQL 5 22 8 3 Updated 6 days ago

stac-server Public
A Node-based STAC API, AWS Serverless
JavaScript MIT 9 22 28 2 Updated 8 days ago

stactools Public
Command line utility and Python library for STAC
Python 14 28 32 (1 issue needs help) 3 Updated 14 days ago

titiler-pgstac Public
TiTiler + PgSTAC
Python MIT 0 9 0 0 Updated 15 days ago

stac-api-validator Public
A STAC API validation client
Python Apache-2.0 1 3 12 1 Updated 22 days ago

stac-fields Public
A minimal STAC library that contains a list of STAC fields with some metadata (title, unit, prefix) and helper functions..
JavaScript Apache-2.0 2 5 8 (1 issue needs help) 0 Updated 29 days ago

stac-migrate Public
A tool to migrate Items, Catalogs and Collections from old versions to the most recent one.
JavaScript Apache-2.0 0 3 2 (1 issue needs help) 0 Updated on Aug 22

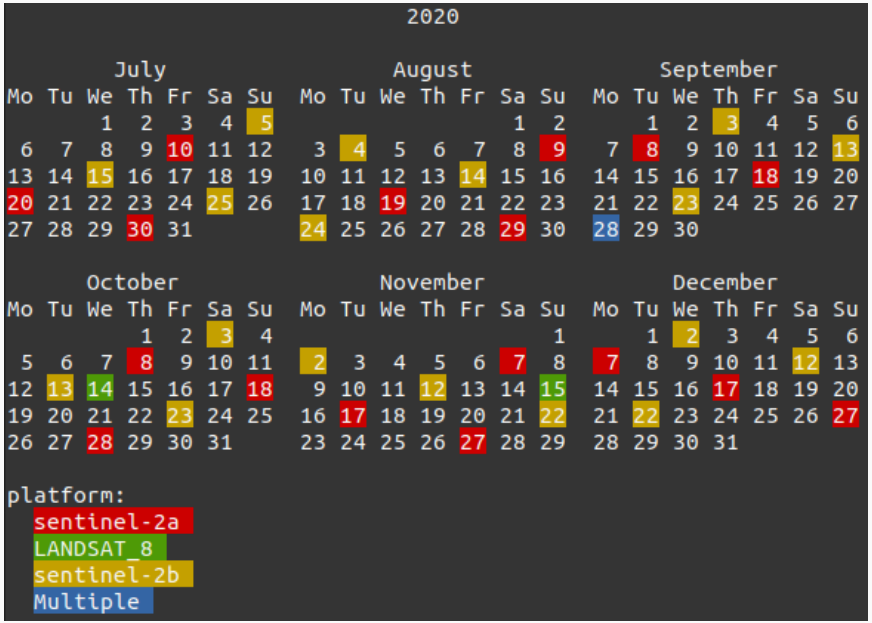
stac-pydantic Public
Pydantic data models for the STAC spec
Python MIT 10 22 18 4 Updated on Aug 18

stac-terminal Public
Output info on STAC Items in the terminal
Python 0 4 0 0 Updated on Aug 13

Element 84

pystac-client

```
$ export STAC_API_URL=https://earth-search.aws.element84.com/v0
$ stac-client search --intersects aoi.json --datetime 2020-07-01/2020-12-31 -c sentinel-s2-l2a-cogs landsat-8-l1-c1 | stacterm cal --label platform
```



stactools

High level CLI/lib for generating and working with STAC

To install all available sub-packages, use

```
pip install stactools[all]
```

`stactools` Features ¶

Use `stac --help` to navigate the varios commands.

- `stac copy` copies STACs and optionally assets
- `stac info` and `stac describe` display information about STACs
- `stac layout` will modify the layout of a STAC based on item properties

One of the focuses of `stactools` is to provide an easy way to plug in functionality for different data sources. For example:

- `stac planet convert-order` will convert a Planet order into a STAC.

See the Command Line Interface documentation for more details about each command.

Table of Contents

- API Reference
  - Core

`stactools` is comprised of subpackages that provide library and CLI functionality. Below is a list of available subpackages.

subpackage	description
`stactools_core`	Contains core library functionality that is used across the other projects
`stactools_cli`	Contains the command line interface (cli) for running the `stactools` command
`stactools_aster`	Methods and commands for working with ASTER data
`stactools_corine`	Methods and commands for working with CORINE Land Cover data
`stactools_planet`	Methods and commands for working with planet data
`stactools_landsat`	Methods and commands for working with landsat data
`stactools_browse`	Contains a command for launching stac-browser against a local STAC

STAC Browser v3

Planet Disaster Data

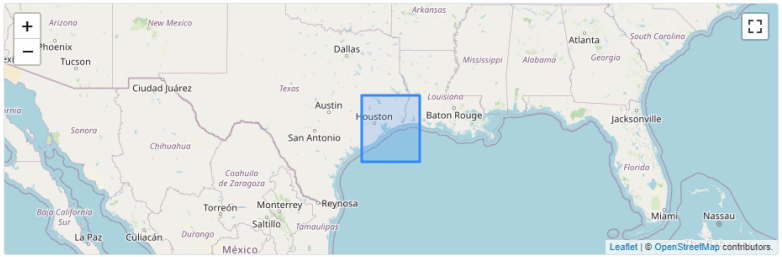
Browse

Introduction

Planet Disaster Data makes imagery available directly to the public, volunteers, humanitarian organizations, and other coordinating bodies in support of the International Charter for Space and Major Disasters. Data is released for individual disaster events, providing a 30 day window pre- and post-disaster. Imagery is provided under Creative Commons licenses, free of charge, with either CC-BY-SA or CC-BY-NC.

disaster open

Licenses	CC-BY-SA-4.0
Temporal Extents	28.8.2017, 18:00:00 UTC until present



Providers

- Planet PRODUCER PROCESSOR
- Planet Disaster Team <disaster-team@planet.com> LICENSOR
- Google Cloud Platform HOST

Catalogs (1)

Hurricane Harvey

Planet Disaster Data makes imagery available directly to the public, volunteers, humanitarian...

Metadata

General

Constellation	1. planetscope 2. skysat
Platform	1. 101c 2. SS02 3. SSC1

Electro-Optical

Gsd	0,9 – 3,7
Cloud Cover	0 % – 24 %

View Geometry

Off-Nadir Angle	0,2 ° – 27,3 °
Sun Elevation	56,3 ° – 65,1 °
Sun Azimuth	122 ° – 231,9 °

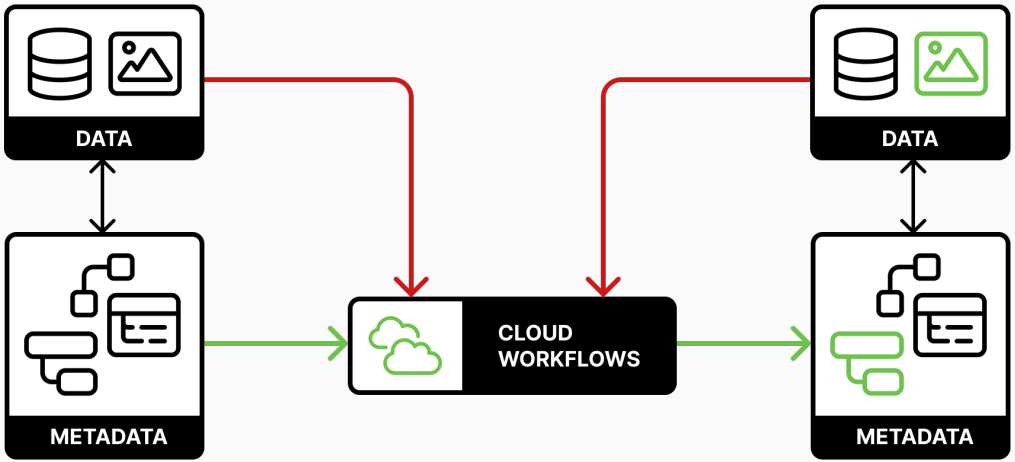
Cloud-Native Geospatial Workflows

Stop thinking of files

```
LC08_L1GT_168005_20190828_20190828_01_RT.tif
LC08_L1GT_168005_20190828_20190828_01_RT_B1.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_B3.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_c10_f1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_cm.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_class_c50_i9.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_tmp_test5.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_ver7.6.tif
LC08_L1GT_168005_20190828_20190828_01_final_class2.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_class.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_final.tif
LC08_L1GT_168005_20190828_20190828_01_test.tif
LC08_L1GT_168005_20190828_20190828_test1.tif
LC08_L1GT_168005_20190828_20190828_test_test.tif
```

Cloud-Native Geospatial Workflows

- Workflows operate on metadata
  - Metadata in, metadata out (STAC In, STAC out)
- Metadata contains complete characterization of data
  - Data location and access



Data provenance

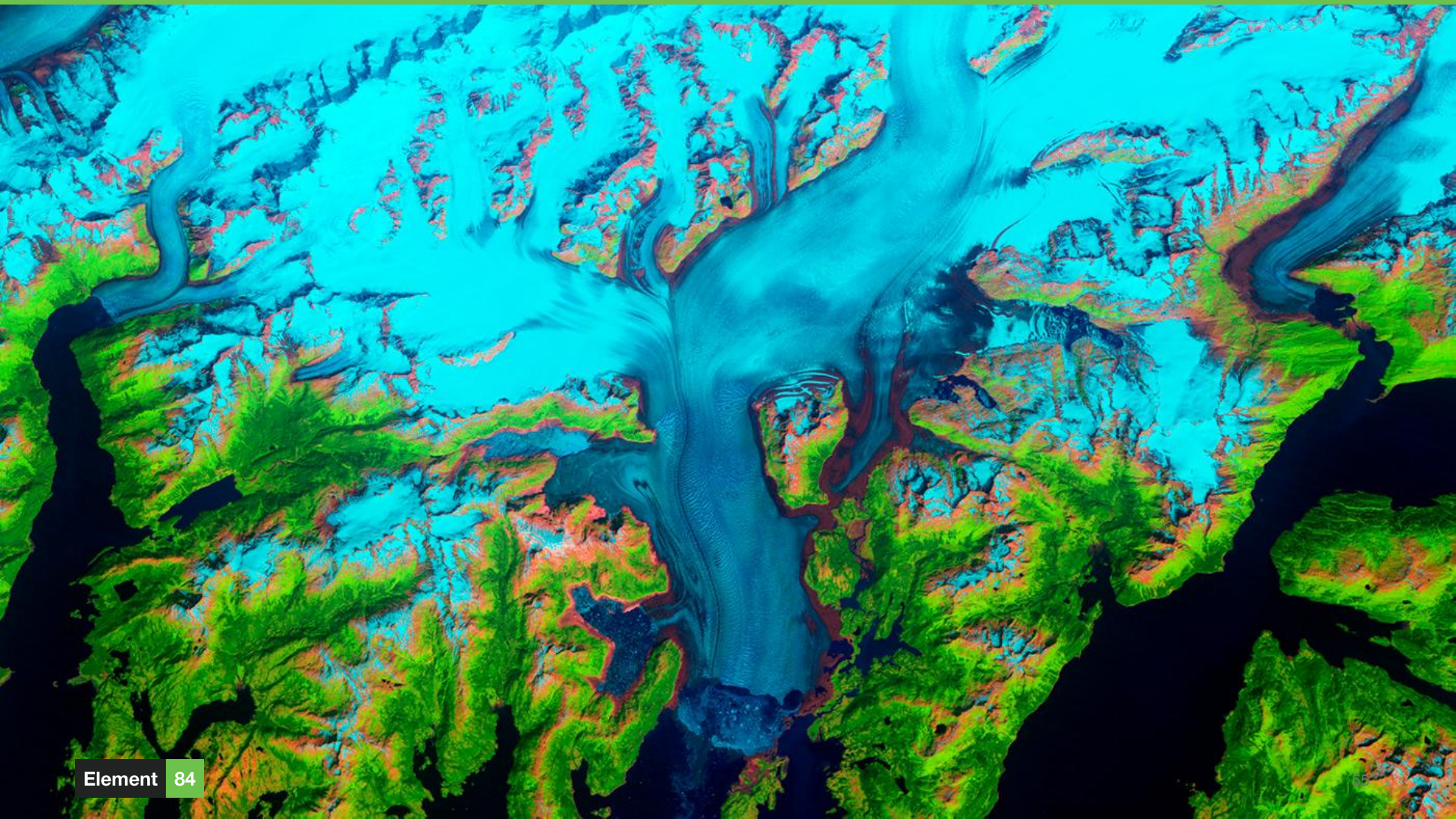

```
"properties": {
  "datetime": "2016-08-22T18:28:23.368922Z",
  "start_datetime": "2016-08-22T18:28:23.368922Z",
  "end_datetime": "2016-08-22T18:28:48.368201Z",
  "created": "2016-08-23T00:38:22Z",
  "platform": "sentinel-1a",
  "constellation": "sentinel-1",
  "mission": "sentinel-1",
  "instruments": [
    "c-sar"
  ],
  "sat:absolute_orbit": 12717,
  "sat:orbit_state": "ascending",
  "sat:relative_orbit": 45,
  "sat:anx_datetime": "2016-08-22T18:24:52.513706Z",
  "sar:instrument_mode": "IW",
  "sar:frequency_band": "C",
  "sar:polarizations": [
    "VV",
    "VH"
  ],
  "sar:product_type": "GRD",
  "processing:lineage": "GRD Post Processing",
  "processing:level": "L1C",
  "processing:facility": "Copernicus S1 Core Ground Segment - DPA",
  "processing:software": {
    "Sentinel-1 IPF": "002.71"
  }
```

```
  "links": [
    {↔},
    {↔},
    {↔},
    {
      "title": "Source STAC Item",
      "rel": "derived_from",
      "href": "https://cirrus-v0-data-1qm7gekzjucbq.s3.us-west-2.amazonaws.com/sentinel-s2-l2a/53/Q/PB/2022/7/S2A_53QPB_20220721_0_L2A/S2A_53QPB_20220721_0_L2A.json",
      "type": "application/json"
    },
    {↔},
    {
      "rel": "parent",
      "href": "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
    },
    {
      "rel": "collection",
      "href": "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
    },
    {
      "rel": "root",
      "href": "https://earth-search.aws.element84.com/v0/"
    }
  ]
},
```

Cloud-Native Geo is the Foundational Architecture for a Queryable Earth

Thank you!

@GeoSkeptic


Element 84